

SCENARIO BASED QUESTION OF CLOUD COMPUTING:

1. Media Company – Cloud Video Streaming Platform

Problem Analysis

The media company must stream videos to **millions of users simultaneously** with **low buffering, high availability, and reasonable cost**.

Proposed Cloud Architecture

- Store videos in **Object Storage** (e.g., Amazon S3, Azure Blob Storage).
- Use a **Content Delivery Network (CDN)** to cache videos near users.
- Place **Load Balancers** in front of streaming servers.
- Deploy streaming servers in containers or virtual machines.
- Enable **Auto Scaling Groups** to add/remove servers automatically.
- Use managed databases for user accounts and watch history.
- Monitor performance using cloud monitoring services.

How Load Balancing and Auto-Scaling Ensure Smooth Streaming

- **Load Balancer**
 - Distributes user requests evenly across servers.
 - Prevents server overload.
 - Redirects traffic if a server fails.
- **Auto Scaling**
 - Adds servers automatically during traffic spikes.
 - Removes extra servers during low traffic.
 - Maintains low latency and minimizes buffering.

Cost Optimization Strategies

- Use **CDN caching** to reduce bandwidth costs.
- Store old videos in low-cost archive storage.
- Use **Spot/Preemptible Instances** for non-critical workloads.
- Enable auto-scaling to avoid paying for idle servers.

- Compress videos using adaptive bitrate streaming.

Conclusion

This architecture provides **high scalability, low latency, reduced buffering, and optimized operational cost.**

2. FinTech Startup – Multi-Cloud Container Deployment

Problem Analysis

The startup needs consistent deployments across multiple cloud providers while ensuring scalability and high availability.

Proposed Solution

- Package applications using **Docker containers**.
- Manage containers using **Kubernetes**.
- Deploy Kubernetes clusters on multiple cloud providers.
- Use Infrastructure as Code (Terraform) for consistent deployments.
- Store container images in a centralized container registry.

Service Discovery and Scaling

- Kubernetes **Service Discovery** automatically finds healthy pods.
- Internal DNS routes requests to available containers.
- **Horizontal Pod Autoscaler (HPA)** increases or decreases pods based on CPU or memory usage.
- Cluster Autoscaler adds worker nodes when required.

Benefits of Multi-Cloud

- Prevents vendor lock-in.
- Provides disaster recovery.
- Improves availability.
- Allows workload distribution across providers.
- Enables cost optimization by selecting cheaper services.

Risks

- Higher management complexity.

- Increased networking cost.
- Security policy differences.
- Data synchronization challenges.
- More difficult monitoring.

Conclusion

Using **Kubernetes with multi-cloud deployment** ensures consistent deployments, scalability, and high availability while reducing dependence on a single cloud provider.

3. E-Learning Platform – Backup and Disaster Recovery

Problem Analysis

Accidental deletion caused data loss. The platform requires fast recovery with minimal downtime.

Proposed Solution

- Enable automatic cloud backups.
- Use object storage **Versioning**.
- Replicate data to another cloud region.
- Schedule database snapshots.
- Use disaster recovery automation.

Achieving RTO and RPO

- **RTO (Recovery Time Objective):**
 - Use automated failover.
 - Keep standby servers ready.
 - Restore services quickly.
- **RPO (Recovery Point Objective):**
 - Perform frequent backups.
 - Enable continuous database replication.
 - Use versioned storage to recover deleted files.

Business Continuity Steps

1. Enable automatic daily backups.
2. Store backup copies in multiple regions.
3. Perform regular disaster recovery testing.
4. Use automated failover mechanisms.
5. Continuously monitor backup status.
6. Maintain documented recovery procedures.

Conclusion

Cloud-native backup, replication, and versioning ensure minimal data loss and rapid service recovery.

4. Government Agency – Hybrid Cloud Architecture

Problem Analysis

Sensitive government data must remain on-premises, while non-sensitive applications can run in the public cloud.

Hybrid Cloud Architecture

- Store confidential data in the on-premises data center.
- Host public-facing applications in the public cloud.
- Connect environments using **VPN** or dedicated private connections.
- Use secure APIs for communication.
- Centralize monitoring and management.

Identity Management

- Implement **Identity and Access Management (IAM)**.
- Enable **Multi-Factor Authentication (MFA)**.
- Apply **Role-Based Access Control (RBAC)**.
- Use Single Sign-On (SSO) across environments.

Encryption

- Encrypt data at rest using AES-256.
- Encrypt data in transit using TLS/SSL.
- Secure encryption keys using Key Management Services.

Challenges

- Managing two infrastructures.
- Compliance and regulatory requirements.
- Increased operational complexity.
- Higher network latency.
- Maintaining consistent security policies.

Conclusion

A hybrid cloud provides flexibility while ensuring compliance, strong security, and controlled access to sensitive government data.

5. SaaS Provider – High Availability and Fault Tolerance

Problem Analysis

The SaaS application must continue operating even if an entire cloud region fails.

Multi-Region Deployment Strategy

- Deploy the application in multiple cloud regions.
- Use Global Load Balancers for traffic routing.
- Maintain active-active or active-passive deployment.
- Replicate databases across regions.
- Store files in geo-redundant storage.

Data Replication and Failover

- Enable synchronous replication within a region.
- Use asynchronous replication between distant regions.
- Configure automatic failover using health checks.
- Redirect traffic instantly to healthy regions during failures.

Monitoring and Alerting

- Monitor CPU, memory, storage, and network usage.
- Track application response time and error rates.
- Configure alerts for failures and unusual activity.
- Use centralized logging dashboards.

- Perform regular disaster recovery drills.

Conclusion

A multi-region cloud architecture with replication, automatic failover, and continuous monitoring ensures **high availability, fault tolerance, and uninterrupted SaaS services** even during regional outages.